Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Quadratic Equations

Learning Objectives

• Define a quadratic equation in standard form.

• Solve quadratics by factorization.

• Solve quadratics by completing the square.

• Apply the quadratic formula.

Key Concepts

• Standard form: ax^2 + bx + c = 0

• Factorization method

• Completing the square

• Quadratic formula: x = (-b +- sqrt(b^2-4ac))/2a

• Word problems leading to quadratics

Teaching Methodology (Step by Step)

1. Show a rectangle-area puzzle that needs x^2.

2. Revise factorization; apply it to quadratics.

3. Demonstrate completing the square geometrically.

4. Derive the quadratic formula once; then use it fluently.

Classroom Activities

• Solve quadratics by all three methods; compare answers.

• Write a word problem that forms a quadratic equation.

Assessment Questions

1. Write the standard form of a quadratic equation.

2. Solve: x^2 - 5x + 6 = 0.

3. Write the quadratic formula.

4. Solve: x^2 + 4x + 4 = 0.

Homework

• Solve the mixed-method exercise.

• Solve five word problems.

Time Allocation

7 periods of 40 minutes each

Unit 2: Theory of Quadratic Equations

Learning Objectives

• Find the discriminant and interpret it.

• Determine the nature of roots.

• Find sum and product of roots.

• Form equations from given roots.

Key Concepts

• Discriminant: b^2 - 4ac

• Nature: real/distinct, equal, imaginary

• Sum = -b/a, Product = c/a

• Forming: x^2 - (sum)x + product = 0

• Symmetric functions of roots

Teaching Methodology (Step by Step)

1. Ask: can we know the roots' nature without solving?
2. Build the discriminant table with examples.
3. Verify sum/product relations with solved equations.
4. Practise forming equations from given roots.

Classroom Activities

• Classify ten quadratics by discriminant without solving.

• Form equations from given root pairs.

Assessment Questions

1. What is the discriminant?
2. Find the nature of roots of x^2 + 2x + 5 = 0.
3. Find sum and product of roots of 2x^2 - 7x + 3 = 0.
4. Form an equation with roots 3 and -2.

Homework

• Complete the discriminant exercise.

• Form equations for five given root pairs.

Time Allocation

6 periods of 40 minutes each

Unit 3: Variations

Learning Objectives

• Define direct and inverse variation.

• Identify joint and combined variation.

• Write variation as an equation with constant k.

• Solve variation word problems.

Key Concepts

• Direct: y = kx

• Inverse: y = k/x

• Joint: y = kxz; Combined variations

• Constant of variation

• Real-life examples: speed-time, workers-days

Teaching Methodology (Step by Step)

1. Use the workers-and-days example: more workers, fewer days.
2. Convert each variation type into its equation.
3. Find k from given data, then answer the question.
4. Give mixed practice to identify the variation type.

Classroom Activities

• Classify ten statements as direct or inverse variation.

• Solve variation problems in pairs.

Assessment Questions

1. Write the equation of direct variation.
2. If y varies inversely as x and y = 6 when x = 4, find k.
3. Give one real-life example of inverse variation.
4. What is joint variation?

Homework

• Solve the variation word-problem exercise.

• Write five real-life examples of each type.

Time Allocation

5 periods of 40 minutes each

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Unit 4: Partial Fractions

Learning Objectives

• Define partial fractions and proper fractions.

• Resolve proper fractions into partial fractions.

• Handle repeated and irreducible quadratic factors.

• Verify results by recombining.

Key Concepts

• Rational vs proper fraction

• Case I: distinct linear factors

• Case II: repeated linear factors

• Case III: irreducible quadratic factor

• Finding constants A, B, C

Teaching Methodology (Step by Step)

1. Present partial fractions as 'un-adding' fractions.
2. Demonstrate Case I fully on the board.
3. Show how repeated factors change the form.
4. Verify by adding the parts back together.

Classroom Activities

• Resolve five fractions of Case I in pairs.

• Verify each answer by recombining.

Assessment Questions

1. What is a proper fraction?
2. Write the partial-fraction form of $1/((x+1)(x+2))$.
3. Resolve: $(3x+1)/((x+1)(x+2))$.
4. How do you verify a partial-fraction answer?

Homework

• Complete the partial-fractions exercise.

• Resolve two fractions of each case.

Time Allocation

5 periods of 40 minutes each

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Unit 5: Sets and Functions

Learning Objectives

• Define set and write sets in roster and set-builder forms.

• Perform union, intersection and complement operations.

• Verify De Morgan's laws with Venn diagrams.

• Define function, domain and range.

Key Concepts

• Set, subset, universal set, empty set

• Roster vs set-builder notation

• Union, intersection, difference, complement

• Venn diagrams; De Morgan's laws

• Function: each input has one output; domain, range

Teaching Methodology (Step by Step)

1. Use the class itself as a set: boys, girls, subsets.
2. Draw Venn diagrams for each operation together.
3. Verify De Morgan's laws with a concrete example.
4. Test relations for the function rule with arrow diagrams.

Classroom Activities

• Draw Venn diagrams for given operations.

• Decide which relations are functions; justify.

Assessment Questions

1. Write {1,2,3} in set-builder form.
2. Find A union B for given sets.
3. State De Morgan's laws.
4. Define a function.

Homework

• Complete the sets exercise with Venn diagrams.

• Find domain and range of five relations.

Time Allocation

6 periods of 40 minutes each

Unit 6: Basic Statistics

Learning Objectives

• Define statistics and its uses.

• Construct frequency tables.

• Draw bar charts, histograms and pie charts.

• Calculate mean, median and mode.

Key Concepts

• Data: raw vs grouped

• Frequency distribution table

• Bar chart, histogram, pie chart

• Mean (average), median, mode

• Choosing the right average

Teaching Methodology (Step by Step)

1. Collect class heights: real data motivates statistics.
2. Build a frequency table from the data together.
3. Draw each graph type from the same data.
4. Calculate the three averages; discuss which fits best.

Classroom Activities

• Survey favourite subjects; present with a pie chart.

• Calculate mean, median, mode of given data sets.

Assessment Questions

1. Define mean, median and mode.
2. Find the mean of 4, 8, 6, 10, 2.
3. Which graph suits parts of a whole?
4. What is a frequency table?

Homework

• Complete the statistics exercise.

• Draw all three graphs for a given data set.

Time Allocation

6 periods of 40 minutes each

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Unit 7: Introduction to Trigonometry

Learning Objectives

• Define trigonometric ratios for an acute angle.

• Find ratios for 30, 45 and 60 degrees.

• Use trigonometric identities.

• Solve height-and-distance problems.

Key Concepts

• sin, cos, tan: opposite/hypotenuse etc.

• Standard angles: 30, 45, 60

• Identities: sin^2 + cos^2 = 1

• Angle of elevation and depression

• Solving right triangles

Teaching Methodology (Step by Step)

1. Draw a right triangle; name sides relative to the angle.
2. Derive the 30-45-60 table with special triangles.
3. Prove the basic identity on the board.
4. Solve a flagpole-height problem as a class.

Classroom Activities

• Measure a tree's height using angle of elevation.

• Complete the standard-angle table from memory.

Assessment Questions

1. Define sin, cos and tan.
2. Write the values of sin 30 and cos 60.
3. Prove: sin^2 A + cos^2 A = 1.
4. What is the angle of elevation?

Homework

• Solve the height-and-distance exercise.

• Prove three trigonometric identities.

Time Allocation

7 periods of 40 minutes each

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Unit 8: Chords of a Circle

Learning Objectives

• State the theorems on chords of a circle.

• Prove that equal chords are equidistant from the centre.

• Prove the perpendicular from the centre bisects the chord.

• Solve riders using chord theorems.

Key Concepts

• Chord, diameter, arc definitions

• Perpendicular from centre bisects the chord

• Equal chords are equidistant from the centre

• Converse theorems

• Riders: applying theorems in proofs

Teaching Methodology (Step by Step)

1. Draw a circle; drop a perpendicular to a chord and measure.
2. Prove the bisection theorem step by step.
3. Prove the equidistant-chords theorem.
4. Guide two riders before independent practice.

Classroom Activities

• Verify the bisection theorem by paper folding.

• Solve riders in pairs; present solutions.

Assessment Questions

1. State the theorem: perpendicular from centre to a chord.
2. What is the longest chord of a circle?
3. State the equal-chords theorem.
4. Define a chord.

Homework

• Write the proofs of the chord theorems.

• Solve three given riders.

Time Allocation

6 periods of 40 minutes each

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